

Beyond the Factor of Safety:

**A CPT-Based Investigation of Liquefaction Triggering and Surface Manifestation
Using the 2011 Christchurch Earthquake Case Histories**

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Independent technical exercise — Python implementation and analysis

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Abstract

This report presents a Python-based implementation of the Boulanger and Idriss (2014) cone penetration test (CPT)-based liquefaction triggering procedure applied to six real case histories from the 2011 Christchurch earthquake. The project had two primary objectives, first, to develop a working implementation of an established liquefaction triggering assessment procedure using Python and second, to investigate whether correcting CPT data for thin-layers and transition-zone effects influences the predicted factor of safety (FS) against liquefaction.

The case histories were selected from the Canterbury CPT-based liquefaction dataset compiled by Geyin et al. (2021). Six CPT soundings were selected to represent the full range of observed 2011 manifestation categories, from no manifestation to severe lateral spreading. The analysis was conducted using both the as-measured CPT parameters and thin-layer-corrected CPT parameters obtained using the inverse filtering procedure of Boulanger and DeJong (2018). For each site, the Boulanger and Idriss (2014) procedure was implemented in Python to calculate stress profiles, soil behavior type index (I_c), estimated fines content, cyclic resistance ratio (CRR), cyclic stress ratio (CSR), and factors of safety against liquefaction triggering.

The results did not show a clear monotonic relationship between the minimum predicted factor of safety and observed manifestation severity. In addition, all six case histories, including the site with no observed liquefaction manifestation, produced minimum factors of safety below 1.0 at some depth. Thin-layer correction generally increased the minimum factor of safety for most sites, with the largest increase observed for the severe manifestation case however, the correction did not eliminate the observed non-monotonic relationship between predicted factor of safety and manifestation severity.

The findings demonstrate the application of Python to the analysis of real-world geotechnical data and provide insight into the behavior and limitations of simplified one-dimensional liquefaction triggering procedures. The project is intended as a focused demonstration of method implementation and engineering understanding rather than a statistically validated assessment of liquefaction prediction accuracy.

1. Introduction

1.1. Project Background

Earthquake-induced liquefaction is a major geotechnical hazard that can cause significant ground deformation and damage to buildings, foundations, roads, buried infrastructure, and other critical facilities. Liquefaction occurs when susceptible, typically loose and saturated soil deposits experience cyclic loading during an earthquake, resulting in the generation of excess pore water pressure and a reduction in effective stress. When the reduction in effective stress is sufficiently large, the soil may experience a substantial loss of strength and stiffness, leading to ground deformation and other surface manifestations.

The 2011 Christchurch earthquake provides an important case history for studying earthquake-induced liquefaction. The earthquake produced widespread liquefaction-related effects across Christchurch, with observed manifestations ranging from no visible manifestation to severe ground deformation and lateral spreading. The event was accompanied by extensive geotechnical investigations, including cone penetration testing (CPT), providing an opportunity to compare analytical liquefaction assessments with observed field behavior. The cone penetration test is widely used in geotechnical engineering for characterizing subsurface soil conditions and assessing liquefaction potential. CPT measurements provide continuous profiles of parameters such as cone tip resistance (q_c) and sleeve friction (f_s), which can be used to estimate soil behavior and evaluate the susceptibility of soil layers to liquefaction triggering. Parameters derived from CPT measurements, including the soil behavior type index (I_c), are commonly incorporated into established liquefaction assessment procedures.

One widely used approach is the CPT-based liquefaction triggering procedure proposed by Boulanger and Idriss (2014). The procedure evaluates the relationship between the cyclic resistance of the soil and the cyclic seismic demand imposed during an earthquake. This relationship is commonly expressed through the cyclic resistance ratio (CRR), cyclic stress ratio (CSR), and factor of safety (FS) against liquefaction triggering. In simplified form, a factor of safety less than 1.0 indicates that the estimated seismic demand exceeds the calculated cyclic resistance at a given depth. Although simplified CPT-based procedures provide valuable tools for engineering assessment, the interpretation of CPT data can be complicated by the presence of thin soil layers and transition zones. Due to the spatial averaging associated with cone penetration, the measured CPT response may not fully capture the true resistance of thin layers. Boulanger and DeJong (2018) proposed an inverse filtering procedure to correct CPT data for these thin-layer and transition-zone effects. Applying such corrections may therefore influence the calculated parameters used in liquefaction triggering assessments and, consequently, the predicted factor of safety.

1.2. Aim

The primary aim of this project is to develop and apply a Python-based implementation of the Boulanger and Idriss (2014) CPT-based liquefaction triggering procedure to six real CPT case histories from the 2011 Christchurch earthquake, and to investigate the influence of thin-layer and transition-zone corrections on the predicted factor of safety against liquefaction triggering.

The project is intended as a focused technical exercise to demonstrate the practical application of an established liquefaction assessment procedure to real-world geotechnical data. It also aims to critically examine the relationship between predicted liquefaction triggering and the observed manifestation severity recorded at the selected Christchurch sites.

1.3. Objectives

The specific objectives of the project are to:

1. Develop a working Python implementation of the Boulanger and Idriss (2014) CPT-based liquefaction triggering procedure, including the calculation of the key parameters required to evaluate liquefaction triggering.
2. Process and analyze real CPT data from six case histories associated with the 2011 Christchurch earthquake using Python-based data processing and numerical analysis techniques.
3. Calculate the soil behavior type index (I_c) and associated liquefaction assessment parameters, including estimated fines content, cyclic resistance ratio (CRR), cyclic stress ratio (CSR), and factor of safety (FS) against liquefaction triggering.
4. Evaluate the effect of thin-layer and transition-zone corrections by performing the liquefaction assessment using both as-measured CPT data and CPT data corrected using the Boulanger and DeJong (2018) inverse filtering procedure.
5. Compare the predicted factor of safety values obtained from the measured and thin-layer-corrected CPT data for each selected case history.
6. Examine the relationship between predicted liquefaction triggering and observed manifestation severity across the six selected sites, ranging from no manifestation to severe lateral spreading.
7. Critically assess the limitations of the simplified liquefaction triggering procedure by examining cases where analytical predictions do not correspond directly with observed field manifestations.

The six case histories are intentionally treated as individual depth-resolved case studies rather than as a statistically representative sample of Christchurch. Consequently, the project does not seek to validate the Boulanger and Idriss (2014) procedure statistically or to estimate the frequency of liquefaction across the Christchurch region. Instead, the analysis focuses on demonstrating the implementation of the method, investigating the influence of CPT thin-layer correction, and developing practical experience in the application of Python to real geotechnical engineering data.

2. Dataset and Site Selection

2.1 Dataset Description

The data used in this study were obtained from the Canterbury CPT-based liquefaction case-history dataset compiled from post-processed geotechnical and earthquake data (Geyin et al., 2021). The curated dataset is available as both a MATLAB structure array and a Python data frame. Each case history contains identifying information, CPT measurements, earthquake-specific parameters, groundwater table depth, and classified liquefaction manifestation information. The CPT data includes depth-dependent measurements of cone tip resistance (q_c), sleeve friction resistance (f_s), and, where available, CPT pore water pressure (u_2). The dataset also provides the pre-drill depth (p_d) for each CPT sounding. In addition to the measured CPT parameters, processed values of cone tip resistance ($q_{c,inv}$) and sleeve friction resistance ($f_{s,inv}$) are provided. According to the dataset documentation, these corrected parameters were processed using the inverse-filtering procedure of Boulanger and DeJong (2018), using the baseline inversion model.

The earthquake-specific information is provided for three Canterbury earthquakes, the 4th September 2010 earthquake ($M_w = 7.1$), the 22nd February 2011 Christchurch earthquake ($M_w = 6.2$), and the 14th February 2016 earthquake ($M_w = 5.7$). Event-specific parameters include earthquake magnitude (M_w), conditional median peak ground acceleration (PGA), PGA uncertainty, groundwater table depth (GWT), and the observed liquefaction manifestation classification.

The liquefaction manifestation classifications were manually assigned to each CPT location based on the observed occurrence and severity of surface manifestations within an approximately 15 m radius of the CPT location. For this study, the analysis was restricted to the 2011 Christchurch earthquake. Therefore, the 2011-specific manifestation classification, groundwater table depth, and conditional median PGA were extracted from the corresponding event-specific arrays in the dataset.

2.2 Data Processing and Filtering

The dataset was processed using Python and the pandas library. The 2011 manifestation classification was first extracted from the manifestation field. Because the dataset contains event-specific information for the 2010, 2011, and 2016 earthquakes, the second element of each event-specific array was used to represent the 2011 earthquake. Case histories with an unknown 2011 manifestation classification, represented by manifestation code 10 per the data manuscript, were excluded from the analysis. The 2011 groundwater table depth was then extracted for the remaining case histories. To avoid using CPT data for which the pre-drill portion extended below the groundwater table, only sites satisfying the condition below were retained:

$$p_d < GWT_{2011}$$

The 2011 conditional median PGA was subsequently extracted, and sites with a PGA less than 0.075g were excluded. This filtering criterion was applied to remove case histories subjected to relatively weak earthquake shaking that were not considered suitable for the intended liquefaction triggering comparison. Finally, the maximum CPT sounding depth was calculated for each remaining case history. Only sites with a maximum sounding depth of at least 6 m were considered eligible for selection. This criterion was imposed to ensure that each selected CPT profile provided sufficient depth coverage for the subsequent liquefaction triggering analysis.

The filtering process can therefore be summarized as follows:

1. Extract the 2011 manifestation classification.
2. Remove sites with unknown manifestation classification (code 10).
3. Extract the 2011 groundwater table depth.
4. Retain sites for which the pre-drill depth is above the 2011 groundwater table.
5. Extract the 2011 conditional median PGA.
6. Retain sites with $\text{PGA} \geq 0.075g$.
7. Retain CPT soundings with a maximum depth of at least 6 m.

2.3 Site Selection

Following the filtering process, one CPT case history was selected from each of the six observed 2011 liquefaction manifestation classes, corresponding to manifestation classes 0 through 5. The six manifestation classes represent increasing levels of observed surface liquefaction effects, ranging from no manifestation to severe lateral spreading. The selection was performed using a Python function that filtered the eligible dataset by manifestation class and randomly selected one CPT sounding from each class. A fixed random seed of 42 was used during the random selection process. This ensured that the selected case histories could be reproduced if the analysis were repeated using the same filtered dataset.

Manifestation Class	Description	CPT ID
0	No manifestation	CPT-HAL-45
1	Minor	ARA-POD11-CPT03
2	Moderate	NNB-POD09-CPT731
3	Severe	NNB-POD09-CPT48
4	Lateral spreading	FND-TC201-CPT12
5	Severe lateral spreading	AVS-POD05-CPT009

Table 1. Selected CPT case histories by observed 2011 liquefaction manifestation class.

The purpose of selecting one case history from each manifestation class was to provide a range of observed field conditions for comparison with the analytical liquefaction triggering predictions. The selected sites were therefore treated as individual case histories rather than as a statistically representative sample of the Christchurch region. The analysis was designed to investigate whether the calculated liquefaction triggering response, expressed primarily through the minimum factor of safety against liquefaction triggering, was consistent with the observed manifestation severity at the selected sites. The study also investigated whether the use of the inverse-filtered CPT parameters, $q_{c,inv}$ and $f_{s,inv}$, produced different liquefaction triggering predictions compared with the original measured CPT parameters, q_c and f_s .

2.4 Measured and Inverse-Filtered CPT Data

Two sets of CPT parameters were considered for each selected case history. The first analysis used the original measured CPT parameters, consisting of cone tip resistance (q_c) and sleeve friction resistance (f_s). These values represent the measured CPT profile provided in the dataset. The second analysis used the inverse-filtered CPT parameters, $q_{c,inv}$ and $f_{s,inv}$. The dataset documentation identifies these parameters as processed using the Boulanger and DeJong (2018) inverse-filtering procedure with the baseline inversion model.

The two datasets were analyzed using the same liquefaction triggering workflow. The results obtained from the measured and inverse-filtered CPT profiles were subsequently compared to evaluate the influence of the CPT processing on the predicted liquefaction triggering response. This approach allowed the study to examine both the relationship between predicted liquefaction triggering and observed manifestation severity and the sensitivity of the results to the use of inverse-filtered CPT measurements.

3. Methodology

3.1 Overview of the Liquefaction Triggering Assessment

The liquefaction triggering assessment was conducted using a Python-based implementation of the CPT-based procedure developed by Boulanger and Idriss (2014). The procedure was applied to six CPT case histories associated with the 2011 Christchurch earthquake, with one site selected from each observed manifestation class ranging from Class 0 (no manifestation) to Class 5 (severe lateral spreading).

The analysis was performed using two sets of CPT measurements for each selected site. The first analysis used the original measured cone tip resistance (q_c) and sleeve friction resistance (f_s). The second analysis used the inverse-filtered CPT parameters ($q_{c,inv}$ and $f_{s,inv}$) provided in the dataset. According to the dataset documentation, these processed CPT parameters were obtained using the Boulanger and DeJong (2018) inverse-filtering procedure with the baseline inversion model.

The liquefaction triggering assessment was based on comparing the cyclic resistance ratio (CRR) of the soil with the cyclic stress ratio (CSR) imposed by the earthquake. The CRR represents the estimated resistance of the soil to earthquake-induced cyclic loading, whereas the CSR represents the estimated cyclic loading demand imposed by the earthquake. The ratio of these two quantities gives the factor of safety (FS) against liquefaction triggering:

$$FS = CRR / CSR$$

A factor of safety less than 1.0 indicates that the estimated cyclic loading demand exceeds the estimated cyclic resistance at that depth, and liquefaction triggering is therefore predicted by the simplified procedure.

The overall computational framework can be summarized as:

$$\text{CPT measurements} \rightarrow \text{Stress profile} \rightarrow I_c \rightarrow FC \rightarrow q_{c,Inv} \rightarrow CRR$$

while the earthquake loading and stress conditions are used to calculate:

$$\text{Earthquake loading} + \text{Stress profile} \rightarrow CSR$$

The two components are then combined through:

$$FS = CRR / CSR$$

The minimum factor of safety calculated over the analyzed CPT profile was extracted for each case history. The minimum values obtained from the measured and inverse-filtered CPT data were then compared across the six observed manifestation classes.

3.2 CPT Data and Stress Profile

3.2.1 CPT Parameters

The liquefaction triggering assessment was based on CPT measurements of cone tip resistance (q_c), sleeve friction resistance (f_s), and pore water pressure (u_2), where available. These parameters provide information about the resistance and response of the subsurface soil during CPT penetration. The cone tip resistance represents the resistance mobilized at the cone tip during penetration, while the sleeve friction represents

the frictional resistance mobilized along the sleeve of the CPT. Together, these measurements provide the basis for characterizing soil behavior and estimating the normalized CPT resistance used in the liquefaction triggering procedure.

The pore pressure measurement, u_2 , was used to obtain the corrected cone tip resistance:

$$q_t = q_c + u_2(1 - a)$$

where:

- q_t = corrected cone tip resistance
- q_c = measured cone tip resistance
- u_2 = pore water pressure measured behind the cone
- a = cone net area ratio

In the Python implementation, a value of $a = 0.80$ was adopted. This value was used in the calculation of q_t for soil behavior type and stress-normalization calculations.

The friction ratio was subsequently calculated as:

$$R_f = (f_s / q_t) \times 100$$

where R_f is expressed as a percentage.

The corrected cone resistance and friction ratio were then used to estimate the soil unit weight and develop the vertical stress profile required for the subsequent liquefaction assessment.

3.2.2 Unit Weight and Total Vertical Stress

The soil unit weight was estimated using the Robertson-style correlation implemented in the Python analysis:

$$\gamma = \gamma_w [0.27 \log_{10}(R_f) + 0.36 \log_{10}(q_t / P_a) + 1.236]$$

where:

- γ = estimated soil unit weight
- γ_w = unit weight of water
- R_f = CPT friction ratio
- q_t = corrected cone tip resistance
- P_a = atmospheric reference pressure

The total vertical stress was then calculated by integrating the estimated unit weight over the depth increments of the CPT profile:

$$\sigma_v(z) = \sum_i \gamma_i \Delta z_i$$

where:

- σ_v = total vertical stress
- γ_i = estimated unit weight of layer i

- Δz_i = thickness of the corresponding depth increment

The total vertical stress represents the total overburden pressure acting on the soil at a given depth.

3.2.3 Pore Water Pressure and Effective Vertical Stress

The groundwater table is an important input to the liquefaction triggering assessment because liquefaction is primarily associated with saturated or nearly saturated soil deposits. In the analysis, hydrostatic pore water pressure was assumed below the groundwater table. The pore water pressure was calculated as:

$$u = 0, \text{ for } z \leq z_{GWT}; \quad u = \gamma_w(z - z_{GWT}), \text{ for } z > z_{GWT}$$

where:

- u = hydrostatic pore water pressure
- z = depth below ground surface
- z_{GWT} = groundwater table depth
- γ_w = unit weight of water

The effective vertical stress was then calculated using:

$$\sigma'_v = \sigma_v - u$$

where σ'_v is the effective vertical stress.

Effective stress is a critical parameter in the liquefaction triggering assessment because it influences both the normalization of CPT resistance and the calculation of the soil's cyclic resistance. As effective stress decreases, the soil's response to cyclic loading and the interpretation of CPT resistance can change significantly. The calculated effective stress profile was therefore used in the subsequent calculation of the soil behavior type index, normalized CPT resistance, cyclic resistance ratio, and cyclic stress ratio.

3.2.4 Numerical Treatment of Stress Values

To avoid numerical instability during the calculations, the Python implementation imposed a minimum effective vertical stress of 1 kPa:

$$\sigma'_v = \max(\sigma'_v, 1.0 \text{ kPa})$$

Similarly, the corrected cone tip resistance was constrained to a minimum value of 1 kPa during subsequent calculations. These constraints were introduced as numerical safeguards in the Python implementation and were not treated as additional physical corrections to the CPT data.

3.3 Fines Content (FC)

The fines content (FC) represents the proportion of fine-grained particles within the soil and is an important parameter in CPT-based liquefaction assessment. The presence of fines can influence the measured cone

penetration resistance and the estimated cyclic resistance of a soil deposit. Consequently, the CPT resistance used in liquefaction triggering calculations is adjusted to account for the influence of fines.

Fines content was estimated from the calculated soil behavior type index (I_c) using the Boulanger and Idriss (2014) functional form with the Christchurch-specific calibration proposed by Lees et al. (2015). Based on approximately 2,600 fines-content measurements from Canterbury Soils, Lees et al. (2015) recommended $C_{FC} = 0.2$. The resulting relationship used in this study was therefore:

$$FC = 80(I_c + C_{FC}) - 137$$

where:

- FC = estimated fines content (%)
- I_c = soil behavior type index
- C_{FC} = calibration parameter

For the Canterbury dataset used in this study, the regional calibration proposed by Lees et al. (2015) was adopted, using $C_{FC} = 0.2$ as reported for Canterbury soils. The fines content used in the present analysis was therefore calculated as:

$$FC = 80(I_c + 0.2) - 137$$

This regional calibration was considered appropriate because the CPT case histories analyzed in this study are in the Canterbury region and form part of the same regional liquefaction case-history database. The calculated fines content was constrained to physically meaningful values between 0% and 100%. This was implemented in Python using:

$$FC = \max(0, \min(100, FC))$$

This constraint ensures that the estimated fines content does not exceed the possible physical range of 0–100%. The estimated fines content was subsequently used to account for the influence of fines on the normalized CPT resistance. Specifically, the estimated FC was incorporated into the fines correction term, Δq_{cIN} , which was then used to obtain the equivalent clean-sand normalized CPT resistance, q_{cINcs} :

$$q_{cINcs} = q_{cIN} + \Delta q_{cIN}$$

where q_{cIN} represents the stress-normalized CPT tip resistance and Δq_{cIN} represents the correction associated with the estimated fines content. Thus, the calculation of fines content provides an important link between the soil behavior classification obtained from I_c and the subsequent estimation of cyclic resistance. In the present study, the workflow can be summarized as:

$$I_c \rightarrow FC \rightarrow \Delta q_{cIN} \rightarrow q_{cINcs} \rightarrow CRR$$

The same Canterbury-calibrated fines-content relationship was applied to both the measured CPT parameters and the inverse-filtered CPT parameters. Therefore, differences in the subsequent liquefaction triggering predictions between the two CPT datasets arise from differences in the underlying CPT measurements and their influence on the calculated soil behavior and normalized CPT resistance.

3.4 Normalized CPT Resistance and Equivalent Clean-Sand Resistance

The CPT cone tip resistance measured in the field is influenced by the effective overburden stress acting at the depth of measurement. Therefore, the measured cone tip resistance cannot be directly compared between soil layers subjected to different effective stress conditions. In the CPT-based liquefaction triggering procedure, the measured CPT resistance is first normalized with respect to effective vertical stress and then adjusted for the influence of fines content.

The normalized CPT resistance was calculated using:

$$q_{cIN} = C_N (q_c / P_a)$$

where:

- q_{cIN} = normalized cone tip resistance
- q_c = measured cone tip resistance
- P_a = atmospheric reference pressure
- C_N = cone resistance normalization factor

The normalization factor C_N was calculated as:

$$C_N = (P_a / \sigma'_v)^m$$

where:

- σ'_v = effective vertical stress
- m = stress exponent

The stress exponent was calculated iteratively as:

$$m = 1.338 - 0.249(q_{cINcs})^{0.264}$$

The calculated value of C_N was limited to a maximum value of 1.7:

$$C_N \leq 1.7$$

This limitation prevents excessive stress normalization at shallow depths or under low effective stress conditions. The calculation of q_{cIN} depends on the stress exponent m , while m itself depends on q_{cINcs} . Consequently, the calculation of the normalized CPT resistance and the equivalent clean-sand resistance was performed iteratively. The iterative procedure was initialized with $q_{cINcs} = 80$. At each iteration, the stress exponent m was calculated first, followed by C_N and q_{cIN} . The effect of fines content was then incorporated through the fines correction term:

$$\Delta q_{cIN} = (11.9 + q_{cIN} / 14.6) \times \exp [1.63 - 9.7 / (FC + 2) - (15.7 / (FC + 2))^2]$$

where:

- Δq_{cIN} = correction to normalized CPT resistance for the influence of fines
- FC = estimated fines content
- q_{cIN} = normalized cone tip resistance

The equivalent clean-sand normalized CPT resistance was subsequently calculated as:

$$q_{cINcs} = q_{cIN} + \Delta q_{cIN}$$

The term q_{cINcs} represents the CPT resistance adjusted to an equivalent clean-sand condition. This parameter is used because the CPT resistance of soils containing fines cannot be directly interpreted using the same liquefaction resistance relationship developed for clean sands. The fines correction therefore allows the measured CPT response to be expressed in terms of an equivalent clean-sand resistance for use in the subsequent cyclic resistance calculation. The calculation was repeated until convergence was achieved.

In the Python implementation, the iteration was stopped when the maximum absolute difference between successive values of q_{cINcs} was less than the specified tolerance of 0.05, or when the maximum number of 20 iterations was reached. The calculated q_{cINcs} was also constrained to a range of 1 to 300 to prevent unrealistic numerical values from affecting the subsequent calculations:

$$1 \leq q_{cINcs} \leq 300$$

The complete normalization and fines-correction process can therefore be summarized as:

$$q_c \rightarrow q_{cIN} \rightarrow \Delta q_{cIN} \rightarrow q_{cINcs}$$

where the first step accounts primarily for the influence of effective overburden stress, while the second adjusts the normalized CPT resistance for the estimated influence of fines. The resulting q_{cINcs} values were then used as the principal CPT-based soil resistance parameter in the calculation of the cyclic resistance ratio (CRR).

3.5 Cyclic Resistance Ratio (CRR)

The cyclic resistance ratio (CRR) represents the capacity of the soil to resist earthquake-induced cyclic loading without undergoing liquefaction triggering. In the simplified CPT-based liquefaction triggering procedure, the CRR is estimated from the equivalent clean-sand normalized cone resistance and is subsequently adjusted to account for the effects of earthquake magnitude and effective overburden stress.

In this study, the CRR was calculated using the equivalent clean-sand normalized cone resistance, q_{cINcs} , obtained from the CPT normalization and fines-content correction described in Section 3.4. The calculation was performed in three main stages: determination of the reference cyclic resistance ratio for a moment magnitude of 7.5, application of a magnitude scaling factor (MSF), and application of the overburden correction factor (K_σ). The final cyclic resistance ratio was calculated as:

$$CRR = CRR_{7.5} \times MSF \times K_\sigma$$

where:

- CRR = cyclic resistance ratio for the earthquake considered
- $CRR_{7.5}$ = reference cyclic resistance ratio corresponding to a moment magnitude of 7.5
- MSF = magnitude scaling factor
- K_σ = overburden stress correction factor

3.5.1 Reference Cyclic Resistance Ratio (CRR_{7.5})

The reference cyclic resistance ratio, $CRR_{7.5}$, was calculated from q_{cINcs} using the deterministic CPT-based relationship implemented in the analysis:

$$CRR_{7.5} = \exp [qc_{1Ncs}/113 + (q_{c1Ncs}/1000)^2 - (q_{c1Ncs}/140)^3 + (q_{c1Ncs}/137)^4 - 2.8]$$

The calculated value was limited to a maximum of 0.6 in the Python implementation. The $CRR_{7.5}$ value represents the estimated cyclic resistance of the soil under a reference earthquake with a moment magnitude of 7.5. Because the actual earthquake considered in this study had a different magnitude, the reference resistance was subsequently modified using the magnitude scaling factor.

3.5.2 Magnitude Scaling Factor (MSF)

The magnitude scaling factor was introduced to account for the influence of earthquake magnitude on liquefaction triggering. Earthquakes of different magnitudes produce different durations and numbers of significant cyclic loading applications. Consequently, the cyclic resistance associated with a reference magnitude of 7.5 must be adjusted when assessing an earthquake of a different magnitude. The MSF used in the present analysis was calculated as:

$$MSF = 1 + (MSF_{max} - 1)[8.64 \exp(-M/4) - 1.325]$$

where:

- MSF = magnitude scaling factor
- MSF_{max} = maximum magnitude scaling factor
- M = moment magnitude of the earthquake

The maximum magnitude scaling factor was calculated from the equivalent clean-sand normalized CPT resistance as:

$$MSF_{max} = 1.09 + (q_{c1Ncs}/180)^3$$

with an upper limit of 2.2. The resulting MSF was applied to the reference $CRR_{7.5}$ to account for the magnitude of the 2011 Christchurch earthquake.

3.5.3 Overburden Stress Correction Factor (K_σ)

The cyclic resistance of a soil is also influenced by the effective overburden stress acting on the soil layer. The overburden correction factor, K_σ , was therefore applied to account for the influence of effective vertical stress on cyclic resistance.

The parameter C_σ was first calculated as:

$$C_\sigma = 1 / [37.3 - 8.27(q_{c1Ncs})^{0.264}]$$

The value of C_σ was limited to a maximum of 0.3. In the Python implementation, this was achieved by limiting the value of q_{c1Ncs} used in the calculation to 211.

The overburden correction factor was then calculated as:

$$K_\sigma = 1 - C_\sigma \ln(\sigma'_v / P_a)$$

where:

- K_σ = overburden stress correction factor
- C_σ = stress correction coefficient
- σ'_v = effective vertical stress
- P_a = atmospheric reference pressure

The calculated K_σ was limited to a maximum value of 1.1.

The application of K_σ accounts for the influence of effective overburden stress on the cyclic resistance of the soil. This correction is particularly relevant because the CPT profiles analyzed in this study extend over different depths and therefore experience varying effective stress conditions.

3.5.4 Final Cyclic Resistance Ratio

The final cyclic resistance ratio was obtained by combining the reference cyclic resistance, magnitude scaling factor, and overburden correction factor:

$$CRR = CRR_{7.5} \times MSF \times K_\sigma$$

The calculated CRR therefore represents the estimated resistance of the soil to the cyclic loading associated with the 2011 Christchurch earthquake at each depth along the CPT profile. The CRR was calculated independently for the measured CPT data and the inverse-filtered CPT data. This allowed the influence of CPT data processing on the estimated cyclic resistance to be investigated. The resulting CRR profiles were subsequently compared with the corresponding cyclic stress ratio (CSR) profiles to determine the factor of safety against liquefaction triggering.

3.6 Cyclic Stress Ratio (CSR)

The cyclic stress ratio (CSR) represents the cyclic shear stress demand imposed on a soil layer during an earthquake. While the cyclic resistance ratio (CRR) describes the capacity of the soil to resist earthquake-induced cyclic loading, the CSR represents the level of cyclic loading generated by the earthquake. Comparing these two quantities provides the basis for evaluating the potential for liquefaction triggering. In this study, the CSR was calculated using the simplified procedure implemented in the Boulanger and Idriss (2014) framework. The calculation considers the earthquake peak ground acceleration (PGA), the total vertical stress, the effective vertical stress, the earthquake magnitude, and the depth-dependent shear stress reduction coefficient, r_d .

The cyclic stress ratio was calculated as:

$$CSR = 0.65 (PGA/g) (\sigma_v/\sigma'_v) r_d$$

where:

- CSR = cyclic stress ratio
- PGA = peak ground acceleration
- g = acceleration due to gravity

- σ_v = total vertical stress
- σ'_v = effective vertical stress
- r_d = shear stress reduction coefficient

In the present analysis, PGA was expressed as a fraction of gravitational acceleration (g). Therefore, the acceleration term used in the CSR calculation is dimensionless.

3.6.1 Total and Effective Vertical Stress

The total vertical stress, σ_v , represents the vertical stress imposed by the overlying soil column. It was calculated by integrating the estimated unit weight of the soil with depth:

$$\sigma_v(z) = \int_0^z \gamma(z) dz$$

where γ is the estimated soil unit weight and z is depth below ground surface. For the discrete CPT measurements used in the analysis, the total vertical stress was calculated incrementally between adjacent depth measurements. The unit weight was estimated using a Robertson-style CPT correlation based on the measured cone resistance and friction ratio. The effective vertical stress was then calculated as:

$$\sigma'_v = \sigma_v - u$$

where u is the pore-water pressure, estimated assuming hydrostatic conditions below the groundwater table:

$$u = 0, \text{ for } z \leq \text{GWT}; \quad u = \gamma_w(z - \text{GWT}), \text{ for } z > \text{GWT}$$

Thus, the effective vertical stress represents the portion of the total vertical stress carried by the soil skeleton. It is particularly important in liquefaction assessment because the resistance of the soil and the normalized CPT parameters are strongly influenced by effective stress.

3.6.2 Peak Ground Acceleration (PGA)

Peak ground acceleration represents the maximum acceleration experienced at the ground surface during the earthquake. It is a key parameter in estimating the cyclic shear stress induced in the soil deposit. For each selected CPT site, the PGA corresponding to the 2011 Christchurch earthquake was extracted from the dataset and used as the earthquake loading parameter. To ensure that the selected sites experienced sufficient earthquake shaking, sites with a 2011 PGA below 0.075g were excluded during the site-selection stage. The PGA values of the six selected sites were subsequently used individually in the CSR calculation. Therefore, the CSR calculation accounts for the spatial variability of earthquake shaking across the selected CPT sites rather than applying a single PGA value to all six sites.

3.6.3 Shear Stress Reduction Coefficient (r_d)

The coefficient r_d accounts for the reduction in cyclic shear stress with depth. The simplified CSR formulation assumes that earthquake-induced shear stresses are generated by the inertial response of the soil column. However, the shear stress induced by the earthquake does not remain constant with depth. In

this study, r_d was calculated using the magnitude and depth-dependent relationship implemented in the analysis:

$$\alpha = -1.012 - 1.126 \sin(z/11.73 + 5.133)$$

$$\beta = 0.106 + 0.118 \sin(z/11.28 + 5.142)$$

The shear stress reduction coefficient was then calculated as:

$$r_d = \exp(\alpha + \beta M)$$

where z is depth below ground surface and M is the earthquake moment magnitude. This formulation allows the estimated cyclic shear stress demand to vary with both depth and earthquake magnitude.

3.6.4 CSR Calculation

The final CSR at each depth was calculated using:

$$CSR = 0.65 (PGA/g) (\sigma_v'/\sigma'_v) r_d$$

The coefficient 0.65 represents the conversion of the peak cyclic shear stress demand into an equivalent uniform cyclic stress representative of the earthquake loading history. The resulting CSR profile therefore represents the cyclic loading demand imposed on the soil at each depth during the 2011 Christchurch earthquake. The CSR was calculated separately for each of the six selected CPT sites using the site-specific PGA, soil stress profile, groundwater conditions, and earthquake magnitude. Because the stress profile and groundwater conditions are independent of whether measured or inverse-filtered CPT data are used, the CSR calculation was evaluated using the corresponding stress profiles generated for each CPT dataset. The resulting CSR was subsequently compared with the calculated CRR to determine the factor of safety against liquefaction triggering. The overall liquefaction triggering assessment can therefore be represented as the earthquake loading feeding into the demand side of the comparison:

Earthquake Loading → CSR

compared against the resistance side of the comparison:

Soil Resistance → CRR

with the factor of safety against liquefaction triggering defined as:

$$FS = CRR / CSR$$

3.7 Factor of Safety Against Liquefaction Triggering

The potential for liquefaction triggering was evaluated by comparing the cyclic resistance ratio (CRR) of the soil with the cyclic stress ratio (CSR) imposed by the 2011 Christchurch earthquake. The factor of safety against liquefaction triggering was calculated at each depth along the CPT profile as:

$$FS = CRR / CSR$$

where:

- FS = factor of safety against liquefaction triggering
- CRR = cyclic resistance ratio of the soil
- CSR = cyclic stress ratio imposed by the earthquake

The factor of safety provides a direct comparison between the capacity of the soil to resist earthquake-induced cyclic loading and the cyclic loading demand imposed by the earthquake. A factor of safety less than 1.0 indicates that the calculated cyclic stress demand exceeds the estimated cyclic resistance, suggesting a potential for liquefaction triggering. Conversely, a factor of safety greater than 1.0 indicates that the calculated cyclic resistance exceeds the estimated earthquake-induced cyclic stress demand. The factor of safety was calculated continuously with depth for each of the six selected CPT sites representing manifestation classes 0 through 5. The analysis was performed for both the measured CPT data and the inverse-filtered CPT data to examine the influence of CPT data processing on the predicted liquefaction triggering response.

3.7.1 Soil Behavior Type Exclusion Criterion

The soil behavior type index, I_c , was used to identify soil layers for which the CPT-based liquefaction triggering procedure was considered applicable. Following the adopted methodology, calculated factor-of-safety values were excluded from the liquefaction triggering assessment where:

$$I_c > 2.6$$

This criterion was implemented in the Python analysis as:

$$FS = np.where(Ic > 2.6, np.nan, FS)$$

Thus, depths with $I_c > 2.6$ were assigned NaN values and were not considered when determining the minimum factor of safety for each CPT site. This approach prevents soil layers classified as behaviorally non-sandy or less susceptible to liquefaction under the adopted CPT-based procedure from controlling the liquefaction triggering assessment.

3.7.2 Critical Factor of Safety

For each selected CPT site, the minimum factor of safety was extracted from the valid depth intervals in the CPT profile. The minimum value was used as the critical factor of safety for representing the liquefaction triggering potential of the site:

$$FS_{min} = \min(FS)$$

where only depths satisfying the adopted I_c applicability criterion were considered. The minimum factor of safety was calculated separately for the measured CPT resistance and the inverse-filtered CPT resistance. The resulting values were summarized for all six manifestation classes. The critical factor of safety provides a concise measure of the most critical liquefaction-prone layer identified within each CPT profile. This

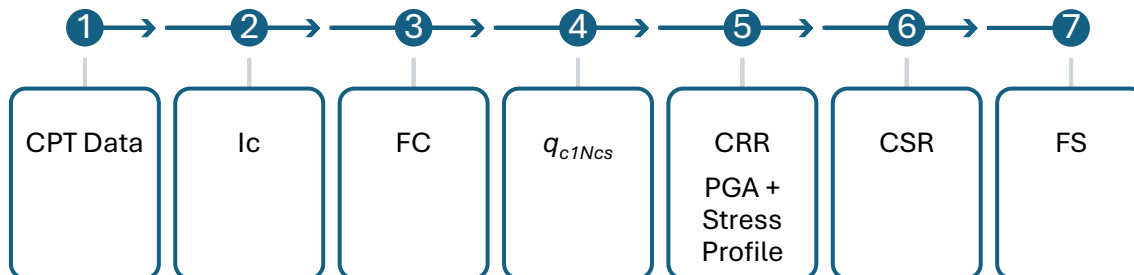
allowed the predicted liquefaction triggering response to be compared directly with the observed manifestation class at the corresponding site.

3.7.3 Comparison with Observed Liquefaction Manifestation

The six selected CPT sites were ordered according to their observed 2011 liquefaction manifestation classes, ranging from Class 0 to Class 5. Class 0 represented the site with no observed liquefaction manifestation, while Classes 1 through 5 represented progressively more severe observed liquefaction effects according to the manifestation classification provided in the dataset. The calculated minimum factors of safety were then compared across the six manifestation classes to investigate whether the CPT-based liquefaction triggering analysis reproduced the observed progression in liquefaction severity. The comparison was performed using two approaches. First, the minimum factor of safety obtained from the measured CPT data was compared with the minimum factor of safety obtained from the inverse-filtered CPT data. Second, the variation in minimum factor of safety was examined across the observed manifestation classes.

The analysis was not intended to establish a statistically representative relationship between liquefaction manifestation severity and factor of safety because only one CPT site was selected for each manifestation class. Instead, the comparison was treated as a focused case-study investigation of whether an established CPT-based liquefaction triggering procedure could provide results that were broadly consistent with observed field performance at selected sites affected by the 2011 Christchurch earthquake.

The overall analytical framework can therefore be summarized as:



The resulting minimum factors of safety were subsequently used as the principal output for comparing the liquefaction triggering predictions with the observed manifestation classes.

4. Results

4.1 Minimum Factor of Safety Results

The minimum factors of safety obtained from the six selected CPT case histories are summarized in Table 2. For each site, the minimum factor of safety was determined from the valid depth intervals along the CPT sounding after excluding layers with $I_c > 2.6$. The analysis was performed using both the measured CPT data (q_c, f_s) and the thin-layer-corrected CPT data ($q_{c,inv}, f_{s,inv}$).

Manifestation Class	Observed 2011 Manifestation	Minimum FS (Measured q_c)	Minimum FS (Corrected $q_{c,inv}$)
0	No manifestation	0.324	0.349
1	Minor	0.310	0.334
2	Moderate	0.259	0.264
3	Severe	0.269	0.347
4	Lateral spreading	0.424	0.450
5	Severe lateral spreading	0.327	0.317

Table 2 Minimum factor of safety by observed 2011 manifestation class.

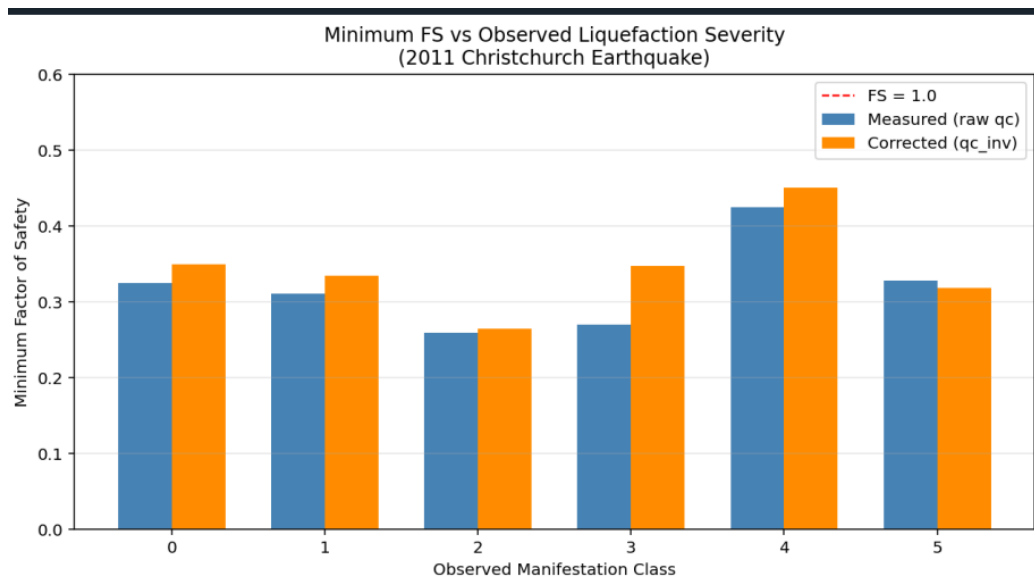


Figure 1 Minimum factor of safety by observed manifestation class, measured vs. thin-layer-corrected CPT data.

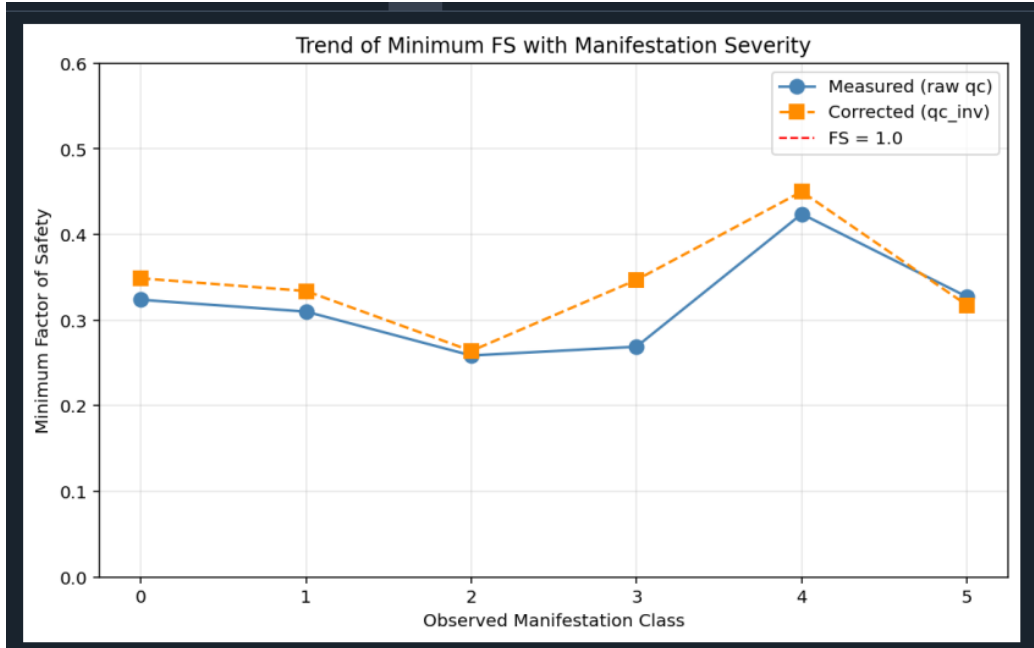


Figure 2 Trend of minimum FS with manifestation severity, shown as a line plot.

The results show that all six selected sites produced minimum factors of safety below 1.0 for both the measured and thin-layer-corrected CPT data. Under the conventional interpretation of the simplified liquefaction triggering framework, this indicates that the calculated cyclic stress demand exceeded the calculated cyclic resistance at least at one depth within each CPT profile. However, the minimum factor of safety does not exhibit a clear monotonic relationship with observed manifestation severity. For the measured CPT data, the minimum FS decreases from 0.324 for Class 0 to 0.259 for Class 2, before increasing to 0.269 for Class 3 and 0.424 for Class 4. The Class 5 site, representing severe lateral spreading, has a minimum FS of 0.327. Notably, the Class 4 site produced the highest minimum FS values (0.424 measured and 0.450 corrected), despite the Class 0 site showing no observed surface manifestation (0.324 measured and 0.349 corrected). These results indicate that, for the six case histories examined, minimum FS alone does not reproduce the observed ranking of manifestation severity. The Class 0 case also demonstrates that an $FS < 1.0$ does not necessarily correspond directly to the occurrence or severity of surface liquefaction manifestation. This highlights the distinction between predicting liquefaction triggering at a particular depth and predicting surface manifestations at the ground surface.

4.2 Effect of Thin-Layer Correction on Predicted Liquefaction Response

The influence of thin-layer correction on the predicted liquefaction response was evaluated by comparing the minimum factors of safety obtained from the measured CPT data with those obtained from the inverse-filtered CPT data. The comparison was performed for each of the six selected sites representing manifestation classes 0 through 5. The results indicate that the effect of the thin-layer correction is site-

dependent. For manifestation classes 0 through 4, the corrected CPT data resulted in higher minimum factors of safety than the corresponding values obtained from the measured CPT data. The Class 5 site showed a slight reduction in the minimum factor of safety following correction.

For Class 0, the minimum factor of safety increased from 0.324 using the measured CPT data to 0.349 using the corrected data. Similarly, Class 1 increased from 0.310 to 0.334, while Class 2 showed only a small increase from 0.259 to 0.264. The largest increase was observed for Class 3, where the minimum factor of safety increased from 0.269 to 0.347. For Class 4, the minimum factor of safety increased from 0.424 to 0.450. In contrast, the Class 5 site showed a decrease in minimum factor of safety from 0.327 for the measured CPT data to 0.317 for the corrected CPT data. This indicates that the thin-layer correction does not produce a uniform change in the calculated liquefaction triggering response across all sites. The absolute changes in minimum factor of safety are summarized in Table 3.

Manifestation Class	Measured FS	Corrected FS	Change in FS
0	0.324	0.349	+0.025
1	0.310	0.334	+0.024
2	0.259	0.264	+0.005
3	0.269	0.347	+0.078
4	0.424	0.450	+0.026
5	0.327	0.317	-0.010

Table 3 Change in minimum factor of safety following thin-layer correction.

The largest change occurred at the Class 3 site, where the minimum factor of safety increased by 0.078 following the correction, while the Class 2 site showed the smallest change, increasing by only 0.005. The Class 5 site was the only case where the correction reduced the minimum factor of safety. Despite these differences, all six sites remained below ($FS=1.0$) for both measured and corrected CPT data. Therefore, the thin-layer correction did not change the overall liquefaction triggering classification of any site. Instead, it modified the magnitude of the predicted factor of safety, with the direction and extent of the change varying between sites. The correction also did not produce a systematic improvement in the relationship between minimum FS and observed manifestation severity. For example, the Class 4 site had the highest corrected minimum FS of 0.450, while the Class 2 site had a much lower value of 0.264. Similarly, the Class 0 site, which showed no observed surface manifestation, had a corrected minimum FS of 0.349.

Overall, the results indicate that the thin-layer correction can influence calculated liquefaction response on a site-specific basis but does not consistently increase or decrease predicted resistance. Its primary effect was therefore quantitative rather than categorical, as all six sites remained below the adopted ($FS=1.0$) triggering criterion.

4.3 Factor of Safety Profiles with Depth

The variation of factor of safety with depth was examined to provide a more detailed interpretation of the liquefaction triggering results beyond the minimum factor of safety reported for each site. Figure 3 presents the factor of safety profiles for all six selected CPT sites using the measured CPT data, while Figure 4 compares the profiles for the Class 0 site, where no liquefaction manifestation was observed, and the Class 5 site, which experienced severe lateral spreading.

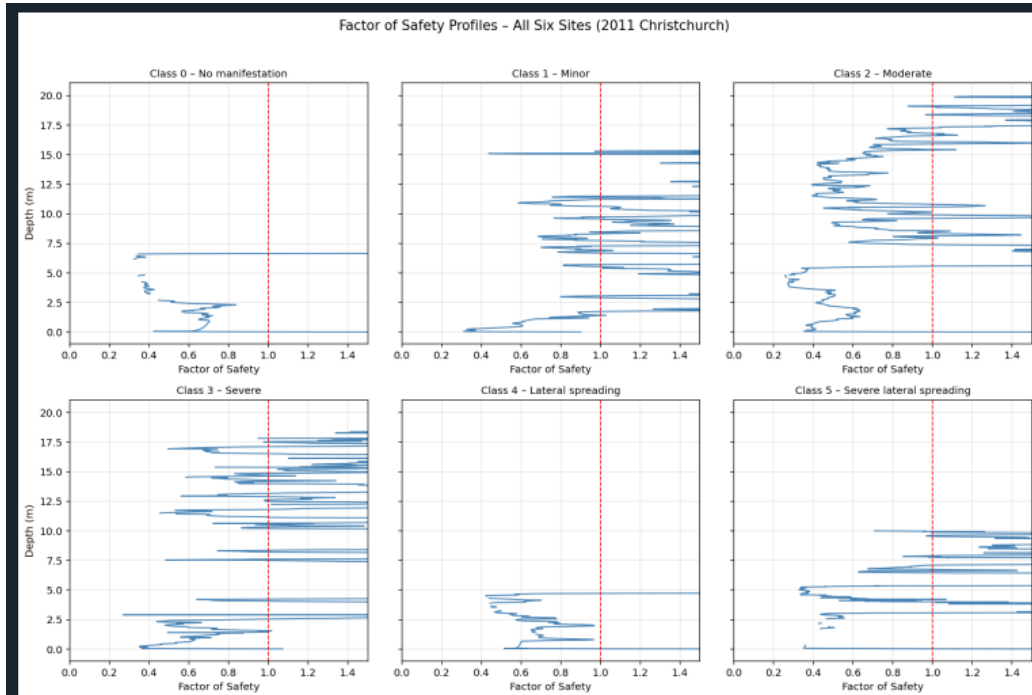


Figure 3 Factor of safety profiles versus depth for all six selected CPT sites (measured data).



Figure 4 Factor of safety profiles versus depth: Class 0 (no manifestation) compared with Class 5 (severe lateral spreading).

The factor of safety profiles provide a depth-resolved representation of liquefaction triggering potential throughout each CPT sounding. Unlike the minimum FS, which represents only the most critical valid depth, the profiles show how the predicted relationship between cyclic resistance and cyclic loading varies with changes in soil properties and stress conditions throughout the soil column.

The (FS=1.0) criterion provides a reference for interpretation, with values below 1.0 indicating depths where the calculated cyclic stress demand exceeds the calculated cyclic resistance according to the adopted simplified triggering procedure. The profiles show that low-FS zones occur at different depths within all six sites, demonstrating that liquefaction triggering potential varies throughout each soil profile.

An important observation is that the presence of low-FS zones does not correspond directly to observed manifestation severity. The Class 0 site, despite having no observed surface manifestation, produced a minimum measured FS of 0.324, while the Class 5 site, which experienced severe lateral spreading, produced a similar minimum FS of 0.327. This demonstrates that a calculated (FS<1.0) at a particular depth should be interpreted as an indicator of potential subsurface liquefaction triggering rather than a direct prediction of surface manifestation severity. The profiles also help explain why minimum FS does not show a monotonic relationship with manifestation class. The critical minimum value may be controlled by a localized weak layer and does not necessarily represent the overall response of the deposit. Factors such as the thickness and continuity of liquefiable layers, non-liquefiable crusts or interlayers, groundwater conditions, and spatial variability may influence whether subsurface triggering develops into observable surface manifestation.

Overall, the factor of safety profiles provide a more complete representation of the liquefaction assessment than minimum FS values alone. They demonstrate that the Boulanger and Idriss (2014) procedure can identify potential triggering zones at depth, but the resulting FS profiles should not be interpreted as direct measures of the severity of surface liquefaction manifestation.

4.4 Key Findings

The analysis of six CPT case histories from the 2011 Christchurch earthquake, representing manifestation classes 0 through 5, produced several key findings.

1. All six selected sites were predicted to have potential for liquefaction triggering at some depth. The minimum factors of safety calculated from both the measured and thin-layer-corrected CPT data were below 1.0 for all six sites. This indicates that the calculated cyclic stress demand exceeded the calculated cyclic resistance at least at one valid depth within each CPT profile.
2. The minimum factor of safety did not show a systematic relationship with observed manifestation severity. The measured minimum FS values ranged from 0.259 to 0.424, while the corrected values

ranged from 0.264 to 0.450. The lowest minimum FS occurred at the Class 2 site rather than at the Class 5 site, which experienced severe lateral spreading. Conversely, the Class 4 site produced the highest minimum FS despite exhibiting lateral spreading.

3. The Class 0 site demonstrated that predicted triggering does not necessarily result in observable surface manifestation. Although no liquefaction manifestation was observed at the Class 0 site, the analysis produced minimum FS values of 0.324 and 0.349 for the measured and corrected CPT data, respectively. This indicates that the triggering procedure predicted potential liquefaction at depth despite the absence of observed surface manifestation.
4. The influence of thin-layer correction was site-dependent. The correction increased the minimum factor of safety for classes 0 through 4 but slightly decreased the minimum factor of safety for Class 5. The largest increase occurred for Class 3, where the minimum FS increased from 0.269 to 0.347. This demonstrates that CPT data processing can influence the quantitative liquefaction triggering prediction, although the direction and magnitude of the effect vary between sites.
5. Thin-layer correction did not change the overall triggering classification of the selected sites. Despite changes in the calculated minimum FS, all six sites remained below $FS = 1.0$ for both the measured and corrected CPT data. Therefore, the correction influenced the magnitude of the predicted factor of safety but did not change the overall conclusion that potential liquefaction triggering was identified at each selected site.
6. Minimum factor of safety alone is insufficient to explain observed liquefaction manifestation severity. The results indicate that the occurrence and severity of surface manifestation cannot be explained solely by the minimum factor of safety obtained from a CPT-based triggering analysis. Other factors, including the thickness and continuity of liquefiable layers, subsurface stratigraphy, groundwater conditions, non-liquefiable crusts or interlayers, and site geometry, may influence whether subsurface liquefaction develops into observable surface manifestations or lateral spreading.

Generally, the six case histories demonstrate that the adopted CPT-based liquefaction triggering procedure can identify potential liquefaction triggering conditions within the investigated soil profiles. However, the calculated minimum factor of safety does not reproduce the observed ranking of manifestation severity across the six sites. The results therefore highlight the distinction between liquefaction triggering, which is evaluated through the comparison of CRR and CSR, and liquefaction manifestation, which represents the resulting ground-surface response.

Given that only one site was selected for each manifestation class, these findings should be interpreted as a focused case-study comparison rather than a statistical validation of the liquefaction triggering procedure. Nevertheless, the analysis provides practical insight into the application of CPT-based liquefaction assessment to real earthquake case histories and demonstrates the importance of considering both subsurface triggering conditions and broader site characteristics when interpreting observed liquefaction manifestations.

5. Discussion

5.1 Interpretation of Liquefaction Triggering Predictions

The results from the six selected CPT case histories show that the Boulanger and Idriss (2014) CPT-based liquefaction triggering procedure identified potential triggering conditions at all sites. Minimum factors of safety from both measured and thin-layer-corrected CPT data were below 1.0, indicating that the calculated cyclic loading demand exceeded the estimated cyclic resistance at least at one susceptible depth at each site.

However, the results demonstrate that liquefaction triggering at depth does not directly correspond to the occurrence or severity of surface manifestation. The Class 0 site, despite having no observed manifestation, produced minimum FS values of 0.324 and 0.349 for measured and corrected data, respectively. This highlights the distinction between triggering within a soil layer and the development of surface manifestations such as ejecta, settlement, cracking, or lateral spreading.

The minimum FS also does not account for factors such as the thickness and continuity of liquefiable layers, non-liquefiable crusts or interlayers, drainage conditions, and pathways for excess pore-water and soil migration. Consequently, a low minimum FS may result from a localized weak layer without producing significant surface manifestation.

This is evident when comparing the Class 0 and Class 5 sites, which produced similar measured minimum FS values of 0.324 and 0.327, respectively, despite substantially different observed responses. Similarly, the Class 4 site, associated with lateral spreading, produced the highest minimum FS values of 0.424 and 0.450. These results reinforce that minimum FS alone cannot capture the complexity of observed ground response.

Lateral spreading is also influenced by geological and geomorphological conditions, such as free faces, slopes, and river channels, which can promote lateral ground deformation following liquefaction. Overall, the results indicate that factor of safety should be interpreted as an indicator of liquefaction triggering potential rather than a direct predictor of liquefaction manifestation severity.

5.2 Interpretation of the Thin-Layer Correction

The comparison between measured and thin-layer-corrected CPT data demonstrates that inverse filtering can influence the calculated liquefaction triggering response. The correction increased the minimum factor of safety for five of the six sites, while the Class 5 site experienced a small decrease.

The largest increase occurred at the Class 3 site, where the minimum factor of safety increased from 0.269 to 0.347. This represents the largest absolute change among the six sites and indicates

that the thin-layer correction had a relatively strong influence on the calculated liquefaction triggering response for this CPT profile.

In contrast, the Class 2 site showed only a small increase in minimum FS, from 0.259 to 0.264. The Class 5 site experienced a decrease from 0.327 to 0.317. These differences indicate that the effect of the correction is dependent on the characteristics of the individual CPT profile rather than producing a uniform adjustment to the liquefaction triggering prediction.

The site-specific behavior is reasonable because the thin-layer correction modifies the interpretation of the CPT resistance in profiles where thin soil layers may be affected by the finite resolution of the CPT cone. The resulting changes in CPT resistance influence the subsequent calculations of soil behavior type, fines content, normalized cone resistance, cyclic resistance ratio, and ultimately the factor of safety.

The results therefore demonstrate that data processing can influence the quantitative outcome of CPT-based liquefaction assessment. However, the correction did not change the overall triggering interpretation of any of the six sites because all minimum factors of safety remained below 1.0.

The correction also did not produce a clearer relationship between predicted minimum FS and observed manifestation class. For example, the Class 4 site retained the highest corrected minimum FS, while the Class 2 site retained the lowest. The Class 0 site also remained below 1.0 despite having no observed manifestation.

Consequently, the thin-layer correction should be viewed as an important component of CPT data interpretation rather than as a mechanism that necessarily improves agreement with observed manifestation severity. Its principal influence in this study was on the calculated magnitude of the factor of safety at individual sites.

5.3 Why Minimum Factor of Safety Does Not Track Manifestation Severity

One of the most important findings of this study is the absence of a monotonic relationship between minimum factor of safety and observed manifestation severity. This result can be understood by considering the difference between the variables represented by the two classifications.

The manifestation class describes the observed ground-surface response following the earthquake. It reflects the combined effects of subsurface liquefaction, soil stratigraphy, groundwater conditions, ground geometry, and deformation mechanisms. Whereas the minimum factor of safety represents the most critical point identified by the simplified triggering calculation within the CPT profile.

These two quantities therefore describe different aspects of earthquake-induced soil response. The minimum factor of safety is influenced primarily by the relationship between cyclic resistance and cyclic loading demand.

The observed manifestation severity, however, depends on a much broader set of conditions. A low FS may indicate that liquefaction is likely to initiate in a particular soil layer, but it does not directly indicate the

thickness of the liquefied zone, the amount of volumetric strain, the degree of pore-pressure redistribution, or the potential for lateral ground movement.

This distinction provides a plausible explanation for the results obtained in the present study. The Class 2 site produced the lowest minimum FS, but this does not mean that it necessarily experienced the greatest ground deformation. Similarly, the Class 4 site produced the highest minimum FS while still being associated with lateral spreading.

The results therefore suggest that a more complete prediction of liquefaction manifestation severity would require additional information beyond the minimum factor of safety. Such information could include the thickness and continuity of liquefiable layers, predicted post-liquefaction volumetric strain, settlement potential, lateral spreading susceptibility, and site-specific geomorphological conditions.

The findings are particularly relevant to the interpretation of real earthquake case histories. The 2011 Christchurch earthquake produced highly variable liquefaction effects across relatively short distances. This spatial variability reflects the strong influence of local geological and geomorphological conditions on liquefaction manifestation.

Therefore, the absence of a direct relationship between minimum FS and manifestation severity in the six selected sites is not unexpected. Rather, it highlights the complexity of translating a one-dimensional CPT-based triggering assessment into a prediction of three-dimensional ground-surface response.

5.4 Implications for CPT-Based Liquefaction Assessment

The results of this study have several implications for the use of CPT-based liquefaction triggering procedures in engineering practice.

First, CPT-based procedures provide an efficient means of evaluating liquefaction triggering potential because they offer continuous measurements of soil resistance with depth. The approach used in this study allowed the liquefaction triggering response to be evaluated across the full depth of each selected CPT profile rather than relying on isolated laboratory samples.

Second, the results demonstrate that the interpretation of CPT data is an important part of the liquefaction assessment process. The comparison between measured and thin-layer-corrected data showed that data processing can influence the calculated factor of safety. This suggests that engineers should consider the characteristics of the CPT profile and the potential influence of thin layers when interpreting CPT resistance for liquefaction assessment.

Third, the results highlight the importance of distinguishing between liquefaction triggering assessment and liquefaction manifestation prediction. A CPT-based triggering analysis can identify soil layers that are susceptible to liquefaction under a specified earthquake loading scenario, but additional analyses may be required to evaluate whether liquefaction will result in significant surface manifestations.

Finally, the study demonstrates the value of using real earthquake case histories when evaluating engineering methods. By applying the procedure to CPT data collected from sites affected by the 2011 Christchurch earthquake, the analysis provides an opportunity to compare calculated triggering potential with actual observed field performance.

The results show that such comparisons can reveal limitations that may not be apparent when considering the factor-of-safety calculation alone. In particular, the difference between calculated triggering potential and observed manifestation severity emphasizes the need for a broader framework when evaluating earthquake-induced ground failure.

5.5 Limitations of the Present Study

Several limitations should be considered when interpreting the results of this study.

The first limitation is the small number of case histories. Only six CPT sites were selected, with one site representing each manifestation class from 0 to 5. The results therefore cannot be considered statistically representative of the wider Canterbury region or the full population of CPT sites affected by the 2011 Christchurch earthquake.

The second limitation is that the analysis focuses primarily on the minimum factor of safety as the principal indicator of liquefaction triggering potential. While the minimum FS is useful for identifying the most critical location within a CPT profile, it does not capture the thickness, continuity, or spatial distribution of potentially liquefiable layers.

The third limitation is that the study evaluates liquefaction triggering rather than directly modeling post-liquefaction deformation or surface manifestation. The observed manifestation classes include effects such as lateral spreading that are controlled by mechanisms beyond the one-dimensional CRR-CSR triggering framework.

Finally, the results depend on the assumptions and empirical correlations incorporated into the adopted methodology, including the CPT-based estimation of soil behavior type, fines content, cyclic resistance, and cyclic stress. These empirical relationships were developed from specific datasets and calibration frameworks and therefore contain inherent uncertainty when applied to individual sites.

Despite these limitations, the study provides a useful case-study evaluation of CPT-based liquefaction triggering analysis using real earthquake data. It demonstrates both the strengths of the method in identifying potential triggering conditions and the limitations of interpreting the resulting minimum factor of safety as a direct predictor of observed liquefaction manifestation severity.

6. Conclusions and Recommendations

6.1 Conclusions

This study evaluated the application of a CPT-based liquefaction triggering assessment to six selected CPT case histories from the 2011 Christchurch earthquake. One site was selected for each observed manifestation class, ranging from Class 0, representing no observed manifestation, to Class 5, representing severe lateral spreading. The analysis applied the Boulanger and Idriss (2014) CPT-based liquefaction triggering procedure to both measured CPT data and thin-layer-corrected CPT data.

The analysis demonstrated that the adopted procedure identified potential liquefaction triggering conditions at all six selected sites. The minimum factors of safety calculated from both the measured and corrected CPT data were below 1.0 for every manifestation class. This indicates that, according to the adopted triggering criterion, the calculated cyclic stress demand exceeded the estimated cyclic resistance at at least one valid depth within each CPT profile.

However, the predicted minimum factor of safety did not show a systematic relationship with the observed manifestation severity. The lowest minimum factor of safety was obtained for the Class 2 site rather than the Class 5 site, while the Class 4 site produced the highest minimum factor of safety despite being associated with lateral spreading. Furthermore, the Class 0 site, where no surface liquefaction manifestation was observed, also produced a minimum factor of safety below 1.0.

These findings demonstrate an important distinction between liquefaction triggering potential and liquefaction manifestation severity. The factor-of-safety approach provides an assessment of whether liquefaction may initiate within susceptible soil layers under a specified earthquake loading condition. It does not, by itself, predict the extent to which liquefaction will develop into observable surface manifestations or lateral ground deformation.

The comparison of the Class 0 and Class 5 sites further illustrates this distinction. Despite representing substantially different observed field responses, the two sites produced similar measured minimum factors of safety. This indicates that the minimum FS alone cannot explain the severity of the observed ground response.

The thin-layer correction also produced site-dependent changes in the predicted factor of safety. The correction increased the minimum FS for classes 0 through 4 but slightly decreased the minimum FS for Class 5. Nevertheless, all six sites remained below $FS = 1.0$ after correction. Therefore, the correction influenced the quantitative magnitude of the predicted factor of safety but did not change the overall triggering interpretation of any of the six selected sites.

This study demonstrates that CPT-based liquefaction triggering analysis can be effectively implemented using real earthquake case-history data and Python-based data processing. The analysis provides a practical framework for processing CPT measurements, calculating stress profiles, evaluating soil behavior type, estimating fines content, determining cyclic resistance and cyclic stress ratios, and calculating factors of safety along the depth of a CPT sounding.

The principal conclusion of the study is that minimum factor of safety is useful for identifying potential liquefaction triggering zones but should not be interpreted as a direct predictor of liquefaction manifestation severity. The observed ground response depends on a broader combination of subsurface, groundwater,

geological, and geomorphological factors that are not fully represented by a one-dimensional triggering analysis.

Because only six sites were considered, with one site representing each manifestation class, the findings should be interpreted as a focused case-study investigation rather than a statistical validation of the Boulanger and Idriss (2014) procedure. Nevertheless, the study provides useful insight into the interpretation of CPT-based liquefaction assessments and demonstrates the importance of comparing analytical predictions with observed earthquake performance.

6.2 Recommendations for Future Work

Based on the findings and limitations of the present study, several areas are recommended for future investigation.

1. Expand the number of CPT case histories. Future studies should consider analyzing a larger number of CPT sites within each manifestation class. Increasing the number of sites would allow the relationship between predicted liquefaction triggering parameters and observed manifestation severity to be evaluated more systematically and statistically.
2. Investigate the influence of liquefiable layer thickness and continuity. Future analysis should consider not only the minimum factor of safety but also the thickness and continuity of layers identified as susceptible to liquefaction. This may provide a better basis for understanding why sites with similar minimum FS values can exhibit substantially different surface manifestations.
3. Incorporate post-liquefaction deformation analyses. The present study focuses primarily on liquefaction triggering. Future work could extend the analysis to estimate liquefaction-induced settlement and lateral spreading. Such analyses may provide a closer connection between calculated subsurface response and observed ground-surface manifestations.
4. Include site-specific geological and geomorphological conditions. Future investigations should incorporate additional site characteristics, including ground slope, proximity to free faces or waterways, groundwater conditions, and the presence of non-liquefiable crusts or interlayers. These factors may help explain the development of lateral spreading and other surface manifestations that cannot be predicted from minimum FS alone.
5. Evaluate alternative liquefaction assessment procedures. The results could be compared with other established CPT-based liquefaction triggering procedures to investigate whether different methods produce different levels of agreement with the observed Christchurch manifestation data. Such comparisons would help assess the sensitivity of the conclusions to the selected analytical framework.
6. Improve the automation and reproducibility of the Python workflow. The Python-based workflow developed in this study could be further developed into a reusable analysis framework capable of processing larger CPT datasets automatically. Future improvements could include automated quality control, batch processing of multiple CPT sites, standardized visualization, uncertainty analysis, and automated generation of summary tables and figures.
7. Investigate uncertainty in liquefaction triggering assessment. Future work should consider the uncertainty associated with CPT measurements, groundwater conditions, earthquake loading parameters,

empirical correlations, fines-content estimation, and the calculation of CRR and CSR. Incorporating uncertainty analysis could provide a more complete understanding of the confidence associated with the predicted factors of safety.

6.3 Final Statement

The 2011 Christchurch earthquake provides an important natural case study for understanding the complexity of earthquake-induced liquefaction. By combining real CPT data with established liquefaction triggering procedures and Python-based data analysis, this study demonstrated how computational tools can be used to investigate the relationship between predicted subsurface liquefaction triggering and observed field performance.

The analysis showed that the adopted procedure consistently identified potential triggering conditions across the six selected sites, but the calculated minimum factor of safety did not reproduce the observed progression of manifestation severity. This finding reinforces the importance of distinguishing between the initiation of liquefaction and the development of surface manifestations.

The study therefore demonstrates both the usefulness and the limitations of CPT-based liquefaction triggering assessment. While the method provides valuable information about the susceptibility of subsurface soil layers to earthquake-induced liquefaction, a comprehensive interpretation of observed ground manifestations requires consideration of the wider geological, hydrological, and geomorphological conditions governing site response.

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